Day 7

CSS Fundamentals

Styling, Layout, Flexbox and Responsive Design

Topics	Tasks	Expected Output
Intro to CSS Inline / Internal / External Selectors and properties Colors and backgrounds Margin and padding Box model Flexbox Basic responsive design	Style the personal profile page Improve the student registration form Create a navigation bar Use Flexbox for cards/sections Make pages responsive	Styled personal profile page Responsive student registration form

Part 1: Topics

1. Introduction to CSS

CSS (Cascading Style Sheets) controls how HTML elements look: colors, fonts, spacing, layout and responsiveness. HTML provides the structure; CSS provides the presentation. A CSS rule has a **selector** and a **declaration block** of property: value pairs.

```
selector {
  property: value;
}

p {
  color: navy;
  font-size: 16px;
}
```

2. Inline, Internal and External CSS

Type	Where written	Use case
Inline	style attribute on a tag	Quick one-off changes; avoid in real projects
Internal	<style> tag inside <head>	Single-page styling
External	Separate .css file linked with <link>	Best practice; reusable across pages

```
<!-- Inline -->
<h1 style="color: red;">Hello</h1>

<!-- Internal -->
<head>
  <style>
    h1 { color: red; }
  </style>
</head>

<!-- External -->
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

Priority order (highest first): inline, then internal/external (the later rule wins when specificity is equal).

3. Selectors and Properties

```
p            { color: #333; }        /* element selector  */
.card        { padding: 16px; }      /* class selector    */
#header      { background: navy; }   /* id selector       */
*            { box-sizing: border-box; } /* universal     */
h1, h2, h3   { font-family: Arial; } /* group selector    */
nav a        { text-decoration: none; } /* descendant     */
a:hover      { color: orange; }      /* pseudo-class      */
input:focus  { outline: 2px solid blue; }
```

Common properties: **color, font-size, font-family, font-weight, text-align, line-height, border, border-radius, width, height, display**.

4. Colors and Backgrounds

```
color: red;                    /* name */
color: #2563eb;                /* hex  */
color: rgb(37, 99, 235);       /* rgb  */
color: rgba(0, 0, 0, 0.5);     /* with transparency */
color: hsl(220, 80%, 50%);     /* hsl  */
```

```
background-color: #f1f5f9;
background-image: url("bg.jpg");
background-size: cover;
background-repeat: no-repeat;
background-position: center;
background: linear-gradient(135deg, #2563eb, #7c3aed);
```

5. Margin and Padding

- **Margin**: space *outside* an element's border (pushes other elements away).
- **Padding**: space *inside* the border, between content and border.

```
margin: 20px;                 /* all sides */
margin: 10px 20px;            /* top/bottom  left/right */
margin: 10px 20px 30px 40px;  /* top right bottom left (clockwise) */
margin: 0 auto;               /* center a block with a set width */
padding: 15px;
padding-left: 10px;
```

6. The Box Model

Every element is a box made of four layers, from inside out: **content, padding, border, margin**.

```
+-------------------- margin --------------------+
|  +----------------- border -----------------+  |
|  |  +------------- padding --------------+  |  |
|  |  |            CONTENT                 |  |  |
|  |  +------------------------------------+  |  |
|  +-----------------------------------------+  |
+-----------------------------------------------+
```

By default (`content-box`), width applies only to the content, so padding and border make the box larger. Use `border-box` so width includes padding and border:

```
* {
  box-sizing: border-box;
}
.box {
  width: 300px;       /* total width stays 300px */
  padding: 20px;
  border: 2px solid #333;
  margin: 10px;
}
```

7. Flexbox

Flexbox arranges items in one dimension (row or column). Set `display: flex` on the **container**; its children become flex items.

Property	Purpose	Common values
flex-direction	Main axis direction	row, column
justify-content	Align along main axis	flex-start, center, space-between, space-around
align-items	Align along cross axis	stretch, center, flex-start, flex-end
flex-wrap	Allow items to wrap	nowrap, wrap
gap	Space between items	10px, 1rem
flex (on item)	Grow/shrink/basis	1, 1 1 250px

```
.container {
  display: flex;
  justify-content: space-between;
```

```
  align-items: center;
  gap: 20px;
  flex-wrap: wrap;
}
.item { flex: 1; }
```

8. Basic Responsive Design

- Add the viewport meta tag in every HTML page.
- Use relative units: %, rem, em, vw instead of fixed px widths.
- Use `max-width` and flexible images.
- Use media queries to change layout at breakpoints (mobile-first approach).

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">

img { max-width: 100%; height: auto; }

/* Mobile first: default styles are for phones */
.cards { display: flex; flex-direction: column; }

/* Tablets and larger */
@media (min-width: 768px) {
  .cards { flex-direction: row; }
}
```

Device	Typical breakpoint
Mobile	up to 600px
Tablet	768px and up
Desktop	1024px and up

Part 2: Tasks (Complete Code)

Create these files in one folder: `profile.html, register.html, style.css`. Both pages share one external stylesheet that contains the navigation bar, cards, form and responsive rules.

Task 1: Shared Stylesheet (style.css)

```
/* ---------- Reset and base ---------- */
* { margin: 0; padding: 0; box-sizing: border-box; }

body {
  font-family: "Segoe UI", Arial, sans-serif;
  line-height: 1.6;
  color: #1e293b;
  background-color: #f1f5f9;
}
a { text-decoration: none; }
img { max-width: 100%; height: auto; }

.container { width: 90%; max-width: 1000px; margin: 0 auto; }

/* ---------- Task 3: Navigation bar ---------- */
.navbar {
  background-color: #1e293b;
  padding: 14px 0;
}
.navbar .container {
  display: flex;
  justify-content: space-between;
  align-items: center;
}
.logo { color: #fff; font-size: 1.4rem; font-weight: bold; }
.nav-links { list-style: none; display: flex; gap: 24px; }
.nav-links a { color: #cbd5e1; padding: 6px 10px; border-radius: 4px; }
.nav-links a:hover,
.nav-links a.active { background-color: #2563eb; color: #fff; }

/* ---------- Task 1: Profile page ---------- */
.hero {
  background: linear-gradient(135deg, #2563eb, #7c3aed);
  color: #fff;
  text-align: center;
  padding: 50px 20px;
}
.avatar {
  width: 140px; height: 140px;
  border-radius: 50%;
  border: 4px solid #fff;
  object-fit: cover;
  margin-bottom: 14px;
}
.hero h1 { font-size: 2rem; }
.hero p  { opacity: 0.9; }

section { padding: 36px 0; }
section h2 {
  margin-bottom: 14px;
  padding-bottom: 6px;
  border-bottom: 3px solid #2563eb;
  display: inline-block;
}
.tag {
  display: inline-block;
  background: #dbeafe; color: #1d4ed8;
  padding: 4px 12px; margin: 4px 4px 4px 0;
  border-radius: 20px; font-size: 0.9rem;
}
```

```
/* ---------- Task 4: Flexbox cards ---------- */
.cards { display: flex; flex-wrap: wrap; gap: 20px; }
.card {
  flex: 1 1 250px;               /* grow, shrink, min width 250px */
  background: #fff;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}
.card h3 { margin-bottom: 8px; color: #2563eb; }

footer {
  background: #1e293b; color: #cbd5e1;
  text-align: center; padding: 18px;
}

/* ---------- Task 2: Registration form ---------- */
.form-box {
  background: #fff;
  max-width: 600px;
  margin: 40px auto;
  padding: 30px;
  border-radius: 10px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.12);
}
.form-box h2 { text-align: center; margin-bottom: 20px; border: none; display: block; }
.form-row { display: flex; gap: 16px; }
.form-group { flex: 1; display: flex; flex-direction: column; margin-bottom: 16px; }
.form-group label { font-weight: 600; margin-bottom: 6px; }
.form-group input,
.form-group select,
.form-group textarea {
  padding: 10px 12px;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  font-size: 1rem;
  font-family: inherit;
}
.form-group input:focus,
.form-group select:focus,
.form-group textarea:focus {
  outline: none;
  border-color: #2563eb;
  box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.25);
}
.radio-group { display: flex; gap: 18px; padding-top: 6px; }
.btn {
  width: 100%;
  padding: 12px;
  background-color: #2563eb; color: #fff;
  border: none; border-radius: 6px;
  font-size: 1.05rem; cursor: pointer;
}
.btn:hover { background-color: #1d4ed8; }

/* ---------- Task 5: Responsive (mobile) ---------- */
@media (max-width: 700px) {
  .navbar .container { flex-direction: column; gap: 10px; }
  .nav-links { gap: 10px; flex-wrap: wrap; justify-content: center; }
  .hero h1 { font-size: 1.5rem; }
  .form-row { flex-direction: column; gap: 0; }
  .form-box { margin: 20px 12px; padding: 20px; }
}
```

Task 1 and 3: Personal Profile Page (profile.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Profile</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <nav class="navbar">
    <div class="container">
      <a href="profile.html" class="logo">MyPortfolio</a>
      <ul class="nav-links">
        <li><a href="profile.html" class="active">Home</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#projects">Projects</a></li>
        <li><a href="register.html">Register</a></li>
      </ul>
    </div>
  </nav>

  <header class="hero">
    <img class="avatar" src="photo.jpg" alt="Profile photo">
    <h1>Your Name</h1>
    <p>Aspiring Web Developer | Student</p>
  </header>

  <section id="about">
    <div class="container">
      <h2>About Me</h2>
      <p>Write a short introduction about yourself, your education
         and your interests here.</p>
      <h2>Skills</h2>
      <p>
        <span class="tag">HTML</span>
        <span class="tag">CSS</span>
        <span class="tag">Git</span>
        <span class="tag">Problem Solving</span>
      </p>
    </div>
  </section>

  <section id="projects">
    <div class="container">
      <h2>Projects</h2>
      <div class="cards">
        <div class="card">
          <h3>Project One</h3>
          <p>Short description of the first project.</p>
        </div>
        <div class="card">
          <h3>Project Two</h3>
          <p>Short description of the second project.</p>
        </div>
        <div class="card">
          <h3>Project Three</h3>
          <p>Short description of the third project.</p>
        </div>
      </div>
    </div>
  </section>

  <footer>&copy; 2026 Your Name. All rights reserved.</footer>
</body>
</html>
```

Task 2: Responsive Student Registration Form (register.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Student Registration</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <nav class="navbar">
    <div class="container">
      <a href="profile.html" class="logo">MyPortfolio</a>
      <ul class="nav-links">
        <li><a href="profile.html">Home</a></li>
        <li><a href="register.html" class="active">Register</a></li>
      </ul>
    </div>
  </nav>

  <div class="form-box">
    <h2>Student Registration</h2>
    <form action="#" method="post">

      <div class="form-row">
        <div class="form-group">
          <label for="fname">First Name</label>
          <input type="text" id="fname" required>
        </div>
        <div class="form-group">
          <label for="lname">Last Name</label>
          <input type="text" id="lname" required>
        </div>
      </div>

      <div class="form-row">
        <div class="form-group">
          <label for="email">Email</label>
          <input type="email" id="email" required>
        </div>
        <div class="form-group">
          <label for="phone">Phone</label>
          <input type="tel" id="phone">
        </div>
      </div>

      <div class="form-row">
        <div class="form-group">
          <label for="dob">Date of Birth</label>
          <input type="date" id="dob">
        </div>
        <div class="form-group">
          <label for="course">Course</label>
          <select id="course">
            <option>Select course</option>
            <option>Computer Science</option>
            <option>Information Technology</option>
            <option>Electronics</option>
          </select>
        </div>
      </div>

      <div class="form-group">
        <label>Gender</label>
        <div class="radio-group">
          <label><input type="radio" name="g"> Male</label>
          <label><input type="radio" name="g"> Female</label>
          <label><input type="radio" name="g"> Other</label>
        </div>
      </div>

      <div class="form-group">
        <label for="addr">Address</label>
        <textarea id="addr" rows="3"></textarea>
      </div>
```

```
      <button type="submit" class="btn">Register</button>
    </form>
  </div>
</body>
</html>
```

Part 3: Expected Output

Styled Personal Profile Page

- Dark navigation bar at the top with a highlighted active link and hover effect.
- Gradient hero section with a circular profile photo, name and title.
- About and Skills sections with rounded skill tags.
- Project cards arranged side by side using Flexbox, with shadows and rounded corners.
- Footer with copyright text.

Responsive Student Registration Form

- Centered white form card with shadow and consistent spacing.
- Fields styled with padding, borders and a blue focus highlight.
- On desktop: two fields per row (name, contact, date/course).
- On mobile (below 700px): fields stack in a single column and the navbar stacks vertically.
- Full-width button with a hover color change.

Responsive Behavior Summary

Element	Desktop	Mobile (max 700px)
Navbar	Logo left, links right (row)	Logo above links (column)
Project cards	3 cards in a row	Cards wrap to 1 per line
Form rows	2 fields per row	1 field per row
Form box	600px max width, centered	Full width with small side margin

Part 4: Testing and Submission Checklist

- Open pages in a browser and resize the window to check the breakpoint.
- Use DevTools (F12) and Toggle Device Toolbar to test phone sizes.
- Confirm the viewport meta tag exists in both HTML files.
- Confirm style.css is linked correctly (check for no 404 in the Network tab).
- Check that all labels are connected to inputs with for and id.
- Take screenshots of desktop and mobile views for submission.

Quick Revision Notes

- Prefer **external CSS** for maintainable code.
- Set `box-sizing: border-box` globally to avoid sizing surprises.
- Margin = outside space, padding = inside space.
- Flexbox: container gets `display: flex`; use justify-content (main axis) and align-items (cross axis).
- Use `flex-wrap` plus `flex: 1 1 250px` for naturally responsive cards.
- Design mobile-first and add media queries for larger screens.